



Hardware Report

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Bosch Future Mobility Challenge

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Pothole Avoidance

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Statement of Purpose

Relevance

This report will review the progress made in the hardware discipline over the past year. It will address objectives that were defined at the beginning of the year and why they were chosen. The hardware portion of the project can often be neglected when working on this project to the profit of focusing exclusively on better algorithms, faster computation times, and more complicated software. However, the topics observed in the present document are essential to the proper function of all the above. In essence, hardware is literally the foundation on top of which the project is being built ; and similarly to the foundations of the houses in which we leave, it is often not paid attention to at first glance.

A good hardware foundation is not only essential to the proper function of the higher-level disciplines, it can often serve to replace and mitigate issues that arise at these higher levels. Issues that would otherwise require an algorithmic solution are no longer required when proper hardware is implemented. Finally, good hardware helps smooth out the development process by providing those who work on the project with reliable and easy-to-use components.

Audience

This report is intended firstly for the BFMC Organizers, to give them an idea of what our hardware progress over the last year has been. We intend this report to be feedback to the competition detailing how we approach the hardware discipline. However, they are not the only people we have in mind when writing this report. Other teams participating, be it in this year's competition or in the future, are also invited to read this report. We hope that by sharing our progress, problems, and solutions in this way, they'll be able to improve their own designs and not fall into the same pitfalls as us. Finally, future members that would like to take up this project after us are encouraged to read this report. It aims to give them an idea of what they are getting into, what progress has been made, what the design and implementation process of this team is as well as what further work needs to be made.

Acknowledgments

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1 Track

1.1 Description

A track to test the car is necessary for anything in the project to produce meaningful results. While simulation testing provides a very flexible means to test out different iterations and improvements, these results cannot be trusted to translate to real-world implementation. The physical world introduces its own set of constraints, disturbances and uncertainties that require an accurate measure with respect to which performance can be evaluated.

This section will provide a detailed account of how our team created the track that we used for testing this year. The different options considered, available approaches and design decisions will all be covered.

1.2 Initial Design Hypothesis

In our previous year's participation in the competition, we neglected building a track at the profit of more extensive testing in the simulator and implementation of algorithmic solutions. This cost us first when we began rushing to build a track for qualification, and even more so when we arrived on the ground in Cluj and were unable to get our vehicle to move. What is more, our core algorithm requires an exact replica of the track, as it is based on an internal model of the track that it compares with the real track. This made our ragged and rushed partial track all the more useless

It was clear from this experience that we needed to dedicate time and resources to building a high quality track that would enable us to not have the same weaknesses as we had last year. It was on top of the priority list when we started the project in the fall.

The approach to the building of the track required taking into account the following constraints and requirements :

Constraints

- 200 CAD budget
- $10m \times 8m$ space available

Requirements

- Exact replica of part of the competition track, with precise measurements.
- Minimum size of track required : $5m \times 5m$
- Modularity : ability to fold up and store as we have no dedicated space
- Durability of material : needs to be durable enough to withstand assembly and disassembly without breaking.
- Weight of material : needs to stay immobile when car accelerates

- Portion of track to accommodate for variety of situations from the competition
- Construction time not to exceed 1 month

Once these constraints and requirements were defined, our hardware team was set to begin the track construction task.

1.3 Encountered Issues

The following section will detail the process that our team went through to build the track. It will follow a dialectic approach (*i.e.* Problem faced - Solution to Problem Faced) because this structure closely replicates the engineering process. It also makes it easier to come back to this report as reference to how hardware problems might be tackled, should anyone reading ever encounter similar challenges.

1.3.1 Choice of Material & Means

The choice of the material out of which we wanted to build the track was the most important one for this whole task. Most of the constraints and requirements concerned this. The budget was the major factor that separated the choice of one material from another. When comparing the means by which the track could be made, a consultation of the prices for having the track printed directly by a printing service quickly eliminated that option from the ones considered. We would have to make the track ourselves.

The hardware team did a quick review of the materials that are available in Montréal and that would meet our requirements and constraints. To compare these materials, a quantitative metric needed to be defined:

- Target Area : $25m^2$
- Target Cost (Material only) : 100\$
- Target Cost per Meter : $\frac{100\$}{25m^2} = 4\$/m^2$

From these metrics, we were able to assess which solution best responded to our needs.

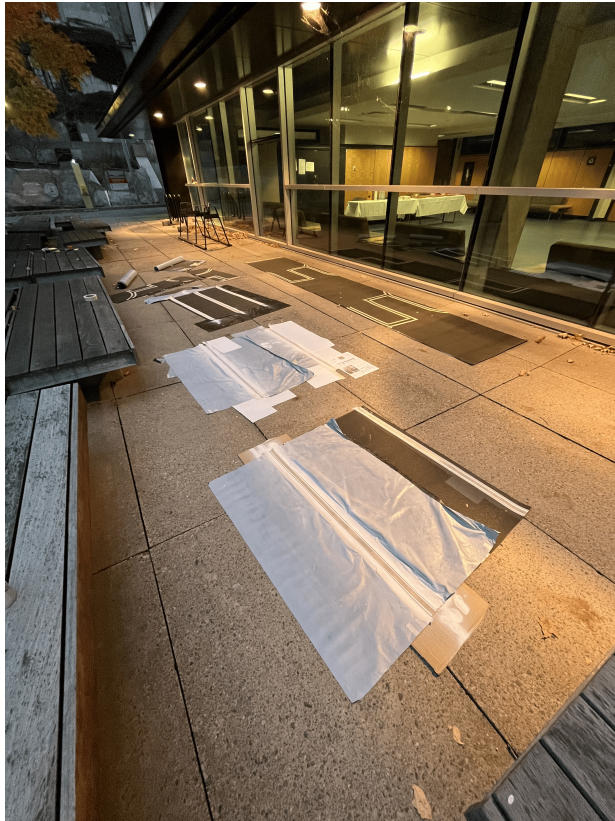
1.3.2 Asphalt Felt & White Paint

The available materials were stored in Table 1. This enabled us to compare the different options that were available. After presenting these options to the team, a concerted decision was made to use Asphalt Felt as the track material. While it was not as inexpensive as the landscaping fabric, it was significantly more durable as it is a material that is usually used in roofing. Compared to canvas or jute, it was significantly less expensive and provided more durability. Asphalt felt is also very similar to a real road and what is being used on the competition ground, which only served to cement our decision.

What was left was the choice of paint and method to apply the paint. We were worried that the nature of the material would make it difficult to find paint that adhered well to

the surface and didn't fade over time. Our first construction of the track therefore saw us use *Inverted Marking Spray Paint*. This type of paint is used on construction sites by land surveyors to indicate different points of interest, and is generally made to be applied on any surface. However, applying the paint proved to be long, tedious given that sections that did not require painting had to be covered with newspapers, and required us to be outside.

When came time to extend the track, we decided to not use the spray paint again but rather to apply paint with a brush or a roller. To choose the paint, we went to Home Depot and ask the paint department to advise us based on our application. They suggested we use a *Stain-Blocking Paint and Primer* applied with a paint roller. This approach proved to be much less tedious and faster than using spray paint.



(a) Spray Painting



(b) Paint Roller

Figure 1: Comparison of Spray Painting and Paint Rolling Techniques

1.3.3 Track Separation

TBD

1.3.4 CAD Modelling

TBD

1.4 Current State

TBD

1.5 Further Development

TBD

2 Appendix

Table 1: Material Cost and Properties

Name	Cost per m ²	Cost for Track	GSM	Width (m)	Color	Unit Sale (m)	Price per Unit(\$)	Weight Per Unit (g)	Details	In store?
Weed Control - Landscaping Fabric	0.90	22	24	0.91	Black	30.48	24.98	682		Yes
Asphalt Felt Type 15	1.03	26	186	0.91	Black	43.89	41.5	7480		Yes
Fiber Glass Reinforced Asphalt Felt	1.84	46	202	0.91	Black	42.89	72.25	7920		Yes
Hardscape Fabric	2.83	71	67	1.83	Grey	15.24	78.98	1870		Yes
Home Décor Fabric - Lentra Black	3.32	83	58	0.90	Black	1.00	2.99	52.2		Yes?
IIC72/ STC 73 Floating Floor Underlayment	3.77	94	135	1.12	*Black	8.31	34.98	1254		No
10 oz Jute Burlap (Natural)	4.52	113	325	1.52	Black	0.91	6.3	452.9	*Price for >25yds	No
10 oz Cotton Duck Canvas (Natural)	4.94	123	330	1.52	Beige	0.91	6.88	459.87	*Price for >25yds	No
Rainbow Felt	5.92	148	198	0.91	Black	1.00	5.39	180.44	*Sale Price + limt. Qty.	Yes
Bulk Broadcloth	6.08	152	100	1.15	Black	27.50	192.23	3162.5		Yes?
10 oz Jute Burlap (Black)	6.46	161	325	1.52	Black	0.91	9	452.9	*Price for >25yds	No
12 oz Jute Burlap (Natural)	7.26	182	400	1.52	Beige	0.91	10.12	557	*Price for >25yds	No
10 oz Cotton Duck Canvas (Black)	8.64	216	330	1.52	Beige	0.91	12.04	459.87	*Price for >25yds	No
Rubber Floor Underlayment - 2mm	9.58	239	947	1.22	*Black	15.24	178	17600		No
Recycled Rubber Runner - 3mm	28.63	716	1176	0.91	Black	0.30	7.98	327.8		Yes

Bibliography